



DATA SCIENCE / EDA PROJECT

EV Purchase Analysis

Understanding the Factors Influencing Electric Vehicle Adoption

A real customer-level survey of 10,000 buyers, analyzed end-to-end from cleaning through a predictive model.

Project Explainer • Interview Preparation Guide



1. Business Problem

An EV company wants to understand which customer characteristics and concerns influence a person's decision to purchase an electric vehicle. By analyzing customer demographics, income, commuting distance, environmental concerns, and range anxiety, the company can identify potential EV buyers and improve its marketing strategy.

“Which customers are most likely to buy an EV, and which factors actually drive that decision versus which ones just seem like they should?”

What this project answers

- How many customers currently intend to buy an EV?
- Does income, age, or demographic profile predict purchase intent?
- Does government subsidy availability change buying behavior?
- Does range anxiety suppress purchase intent, and by how much?
- Does charging-station proximity matter, or just charging feasibility?
- Can we predict, with real accuracy, whether a new customer will buy?

2. Dataset Overview

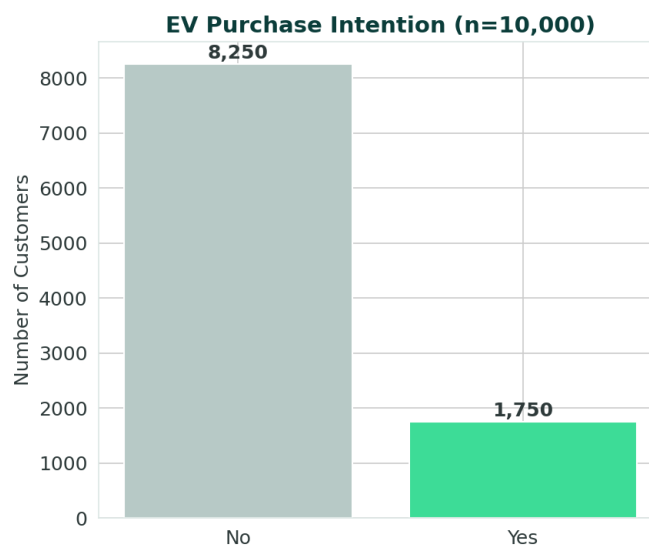
Rows	10,000 customers
Columns	15 (demographics, income, commute, charging access, subsidy, range anxiety, target)
Target	Will_Buy_EV — Yes/No (17.5% positive rate — imbalanced)
Missing values	Income (178), Commute (181), Environmental Concern (184) — filled with median
Duplicates	0; Buyer_ID confirmed unique across all 10,000 rows

3. Data Cleaning

- Filled 178/181/184 missing values (Income, Commute, Environmental Concern) with the column median
- Confirmed 0 duplicate rows and a fully unique Buyer_ID key
- Mapped Will_Buy_EV, Subsidy_Available, and Home_Charging_Possible from Yes/No text to 1/0 for analysis
- Mapped Range_Anxiety_Level (Low/Medium/High) to an ordinal 0/1/2 scale for correlation

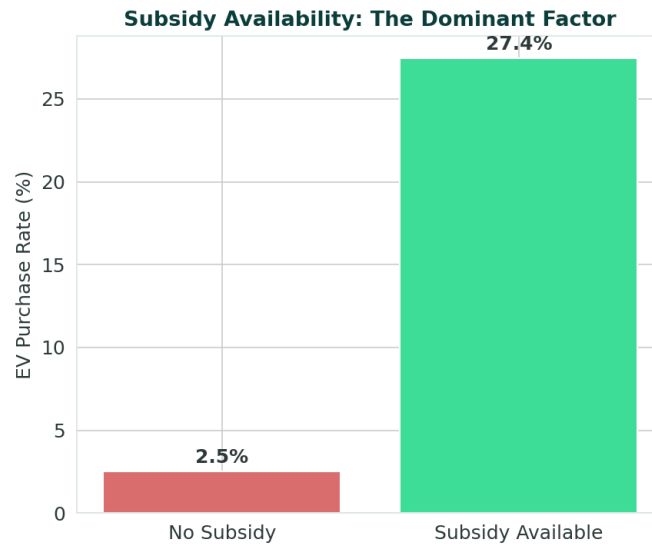
Data-quality note: several columns in this dataset are stored as text (Yes/No, Low/Medium/High) rather than numbers. Averaging or correlating them directly throws an error — they must be explicitly encoded first. This is a common real-world gap between how survey data is stored and how it needs to be shaped for analysis.

4. Exploratory Data Analysis



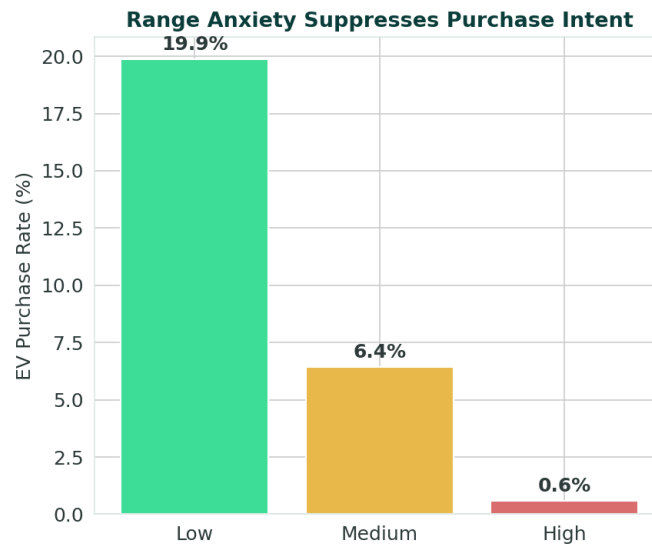
Only 17.5% of surveyed customers currently intend to buy an EV — a meaningfully imbalanced target.

Subsidy availability is the dominant factor



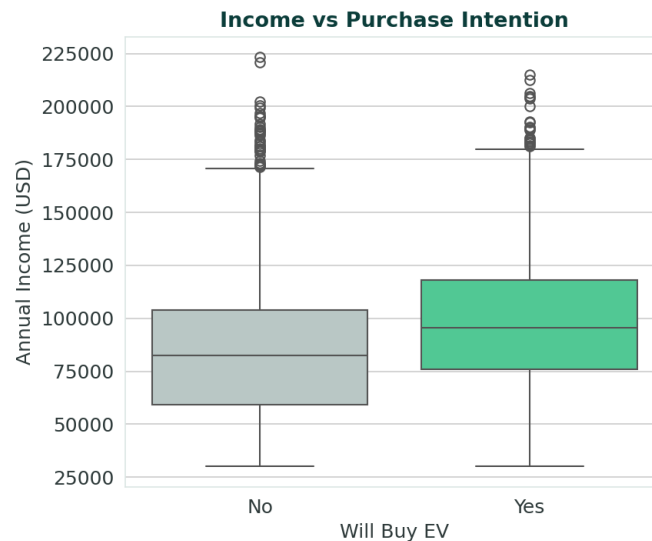
Customers with an available subsidy buy at 27.4%, versus just 2.5% without one — an 11x difference, by far the single strongest factor found anywhere in this dataset.

Range anxiety strongly suppresses intent



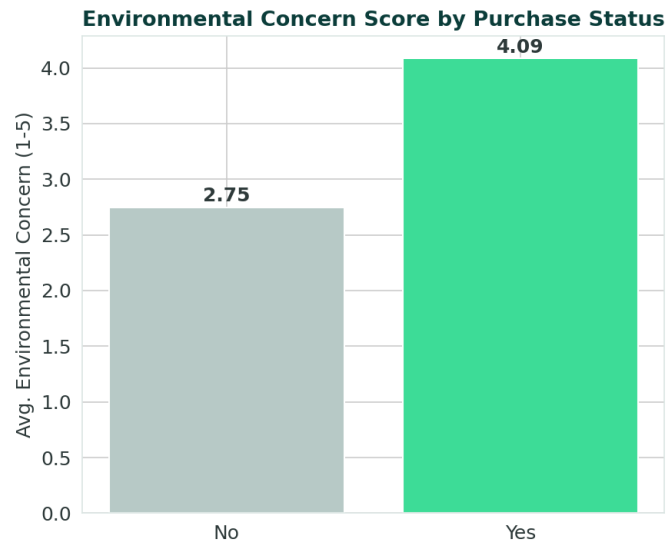
Purchase rate falls from 19.9% (Low anxiety) to 6.4% (Medium) to just 0.6% (High) — range anxiety is close to a hard blocker once it reaches the high end.

Income shows a real, moderate gap



Median income is \$95,716 for buyers versus \$82,612 for non-buyers — a real gap, though smaller in effect than subsidy access or environmental concern.

Environmental concern is the strongest continuous predictor

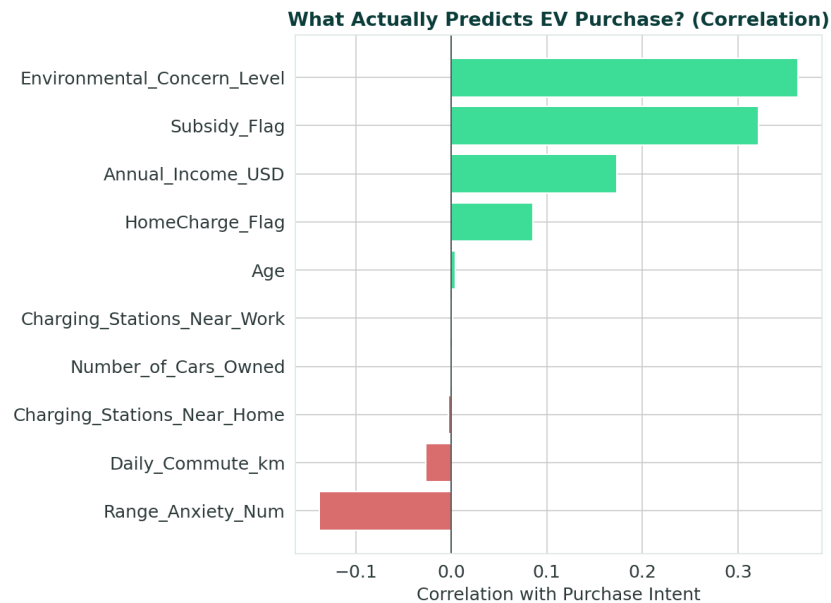


Buyers average 4.09 on a 1-5 environmental concern scale, versus 2.75 for non-buyers — a large, consistent gap.

What doesn't matter (and that's a finding too)

Age, gender, city type, current car type, number of cars owned, daily commute distance, and — notably — the number of charging stations near home or work all showed a negligible relationship with purchase intent (correlations under 0.03 in every case). This runs against the intuitive assumption that nearby charging infrastructure would be a strong driver; the data says otherwise.

Correlation Ranking



Only four factors matter: Environmental Concern, Subsidy, Income, and Range Anxiety. Everything else is noise.

5. Machine Learning

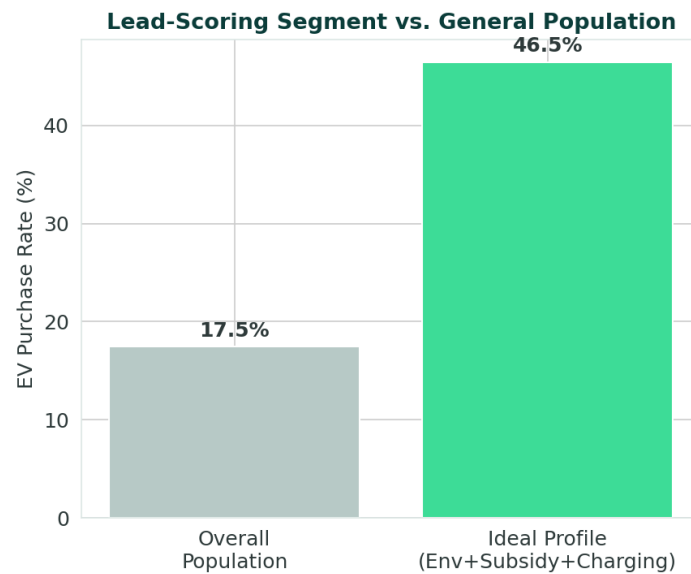
Target: Will_Buy_EV (mapped to 1/0). Features: all columns except Buyer_ID. Categorical fields one-hot encoded. Model: Logistic Regression with **class_weight='balanced'**, since only 17.5% of customers are positive examples.

Metric	Value
Baseline (always predict 'No')	82.5%
Model accuracy	81.4%
Precision	0.48
Recall	0.86
F1-score	0.62
ROC-AUC	0.91

Why accuracy looks worse than the baseline: this is intentional, not a flaw. With class weighting, the model trades a little accuracy for a large gain in recall — it correctly identifies 86% of real buyers, versus 0% for a

model that just predicts “No” every time. ROC-AUC of 0.91 confirms the model separates buyers from non-buyers very well overall; accuracy alone would have hidden that.

6. The Ideal Buyer Profile



Customers who combine above-median environmental concern, subsidy access, and home-charging capability buy at 46.5% — nearly 3x the 17.5% baseline rate — and this segment covers 2,292 of the 10,000 customers (23%). This single rule is the strongest, simplest lead-scoring tool this analysis produced.

7. Business Recommendations

Finding: Subsidy availability produces an 11x swing in purchase rate (2.5% → 27.4%).

Recommendation: Lead marketing and sales conversations with subsidy eligibility — it is the highest-leverage lever available, ahead of any demographic targeting.

Finding: High range anxiety nearly eliminates purchase intent (0.6% buy rate).

Recommendation: Invest in range-anxiety-reducing tools (range calculators, warranty messaging, roadside assistance) before broad demographic campaigns.

Finding: Charging-station proximity showed no measurable effect, despite being an intuitive assumption.

Recommendation: Don't use "chargers per neighborhood" as a targeting signal — the data doesn't support it; charging feasibility (Yes/No) matters, station count doesn't.

8. Your 60-Second Interview Answer

"I analyzed a 10,000-customer EV purchase survey to find out which factors actually predict purchase intent. After cleaning missing values and fixing a common issue where Yes/No and ordinal fields were stored as text, I ranked every feature by correlation with the target. Only four factors mattered: environmental concern, subsidy availability, income, and range anxiety — everything else, including age, gender, and even charging-station proximity, was essentially noise, which corrected an assumption I'd started with.

Subsidy availability turned out to be the single strongest lever, producing an 11x difference in purchase rate.

I trained a Logistic Regression model with class weighting, since only 17.5% of customers were actual buyers, and got a 0.91 ROC-AUC and 86% recall — prioritizing catching real buyers over raw accuracy, which I explained explicitly rather than letting the lower accuracy number speak for itself.

I combined the three strongest factors into a single lead-scoring rule that lifted the buy rate from 17.5% to 46.5% for about a quarter of the customer base — a concrete, immediately actionable output for a sales or marketing team.”

9. Frequently Asked Interview Questions

Q: Why did you use `class_weight='balanced'` instead of just accepting lower recall?

A: With only 17.5% positive examples, an unweighted model tends to just predict the majority class. Class weighting forces it to pay attention to the minority class, trading some accuracy for much better recall — the right trade-off when missing a real buyer is costlier than a false lead.

Q: Why is your model's accuracy lower than the naive baseline?

A: Because the baseline (always predict 'No') scores 82.5% by doing nothing useful. My model deliberately trades a few points of accuracy for a large recall gain (0% → 86%), which I'd always explain rather than let the raw number look worse than it is.

Q: You found charging-station count doesn't matter — doesn't that contradict common sense?

A: It contradicts an intuitive assumption, which is exactly why I checked it rather than assumed it. Whether home charging is feasible at all does matter (Q4); how many stations happen to be nearby doesn't. That distinction itself is a useful, non-obvious finding.

Q: How would you improve this model further?

A: Try Random Forest or Gradient Boosting to capture interaction effects like subsidy × income, test SMOTE as an alternative to class weighting, and validate the lead-scoring rule on a holdout sample before handing it to a sales team.

Q: What's the single most useful output of this project for the business?

A: The three-factor lead-scoring rule (environmental concern + subsidy + home charging) that lifts buy rate from 17.5% to 46.5% for a targetable 23% of customers — simple enough to hand directly to a sales team without needing the underlying model.